

Group #8: Emma Carothers, Daniel Mai, Annalise Dubeck

FYRE Final Project: EDA Umbrella

Lehigh University

Engr 095: Sensing the World module

Fall 2026





Introduction:



Objective

Build an automated system to shield a doghouse from harsh weather.

Who are we helping?

Our dog (named EDA in our prototype) to keep dry

Constraints

- Prone to water damage
- Prone to weather damage
- Structure made of cardboard
- Access to materials

Requirements

- Use homemade PDOT sensor
- At least 2 other inputs
- Build a full prototype
- Provides service or support that can improve quality of life
- Human interaction

Project Materials Overview

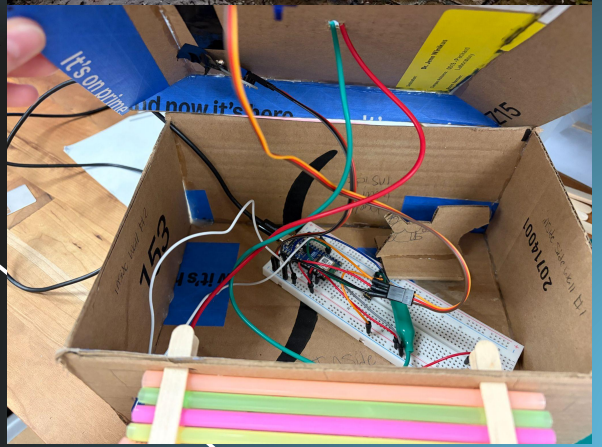
Button	input
Photoresistor light sensor module	input
PDOT moisture sensor	input
Servo motor	output
LED light	output
Cardboard, hot glue, popsicle sticks, plastic straws, duct tape, electrical tape	construction
Breadboard and wires	circuit board
Arduino microcontroller, MicroPython	software

Implementation Overview

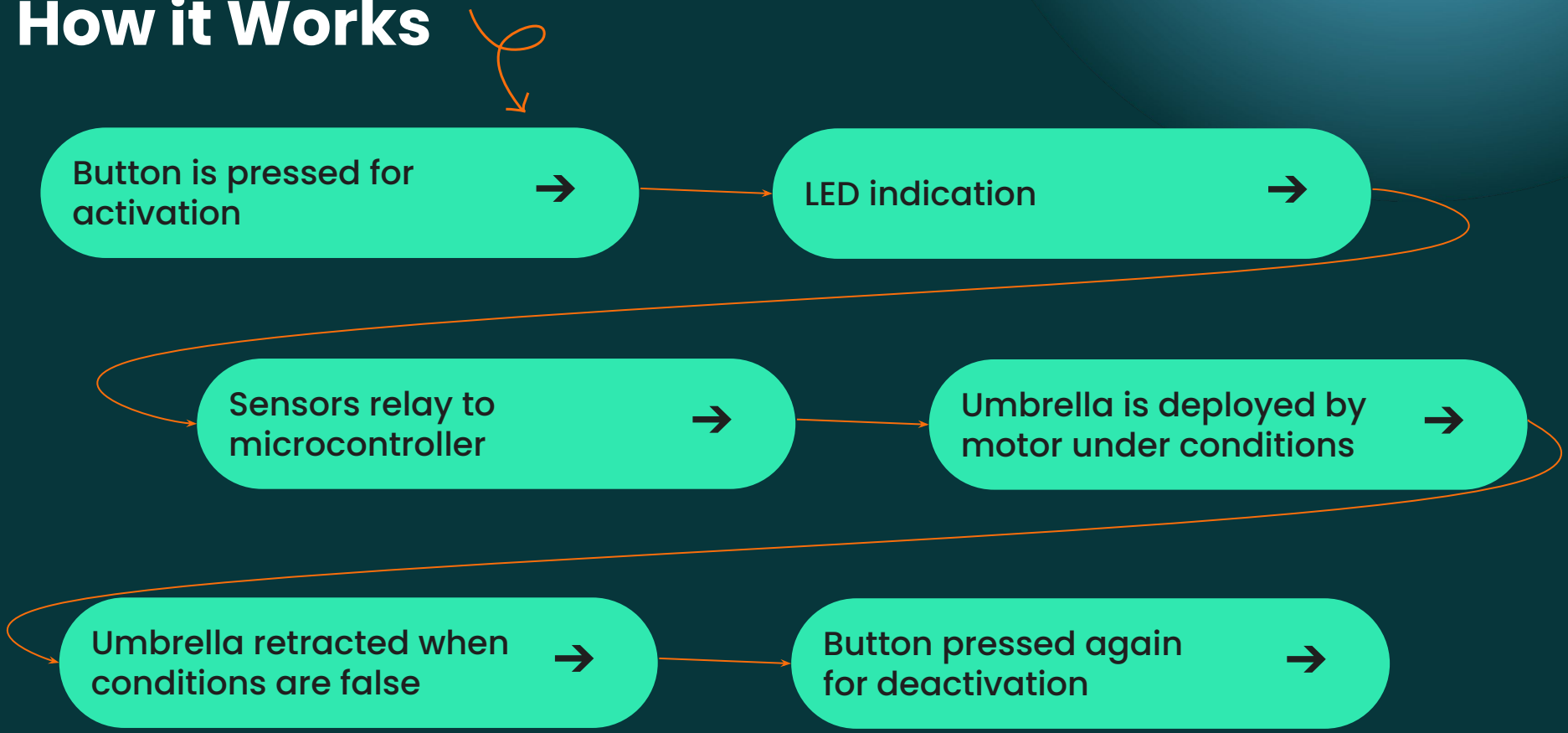
- Building: cardboard house, holes for each component, circuit board inside house
- Circuit: Microcontroller pins connect to one leg of each component
- Use of AI: ChatGPT provided code and pin layout
- Actuating conditions: Rainy AND cloudy

Detailed mock-up





How it Works


Button is pressed for activation
LED indication
Sensors relay to microcontroller
Umbrella is deployed by motor under conditions
Umbrella retracted when conditions are false
Button pressed again for deactivation

Results



- Success!!!
  - Electronics:
    - Uses data from sensors to deploy the umbrella under certain conditions
  - Structure:
    - Covers our dogs and they do not get wet

Conclusion

- Successfully models the automated umbrella
  - Deploys only when *both* moisture and darkness are detected
- Considering current constraints:
  - Final implementation of this prototype would use stronger materials

Demo:

